ASSIGNMENT 08

Submitted By,

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Roll no: 25CEM5R10

Submitted To,

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Thermal,microwave and hyperspectral remote senisng

Plotting of Spectral reflectance curve for NEON Dataset

Introduction to Albedo Spoectrometer mosaic:

An **albedo spectrometer mosaic** is a high-resolution, georeferenced map (or "mosaic") composed of individual data tiles that represent the **albedo** (reflectivity) of a surface as measured by an **imaging spectrometer**.

NEON Data

These products are essential for climate science and environmental monitoring because they allow researchers to track how much solar energy is being absorbed by different landscapes (like forests, snow, or urban areas) over large regions.

Core Components

- **Albedo:** The ratio of solar radiation reflected by a surface to the total amount hitting it. It is expressed as a value between 0 (total absorption) and 1 (total reflection).
- **Imaging Spectrometer:** An instrument that records light in many narrow, contiguous spectral bands. This allows for precise calculations of "spectral albedo" (albedo at specific wavelengths) or "broadband albedo" (integrated across the entire spectrum).
- **Mosaic:** A large-scale image created by stitching together multiple smaller data "flight lines" or "tiles" to provide continuous coverage of a study area.

Production Process

1. **Data Acquisition:** Sensors (often mounted on satellites like **MODIS** or aircraft like **NEON's** Airborne Observation Platform) capture raw reflectance data along flight paths.
2. **Correction:** The data is corrected for atmospheric interference, topography, and the **Bidirectional Reflectance Distribution Function (BRDF)**, which accounts for changes in reflectance based on sun and sensor angles.
3. **Orthorectification:** Each data point is matched to precise geographic coordinates.
4. **Mosaicking:** The processed data is merged into standardized tiles (e.g.,

GeoTIFFs) to create a seamless regional or global map.

Primary Applications

- **Energy Balance Studies:** Characterizing how much energy is absorbed by Earth's surface to understand local and global heating.
- **Vegetation Monitoring:** Tracking seasonal changes in plant health and cover, which significantly alter surface albedo.
- **Climate Feedback Analysis:** Studying the **ice-albedo feedback**, where melting ice lowers albedo, leading to further warming and more melting.

Description:

Total amount of solar radiation in the 0.4 to 2.5 micron band reflected by the Earth's surface into an upward hemisphere divided by the total amount incident from this hemisphere. Data are derived from the "Spectrometer orthorectified surface directional reflectance - flightline" data product (DP1.30006.001) and are mosaicked into 1 km x 1 km geotiff tiles.

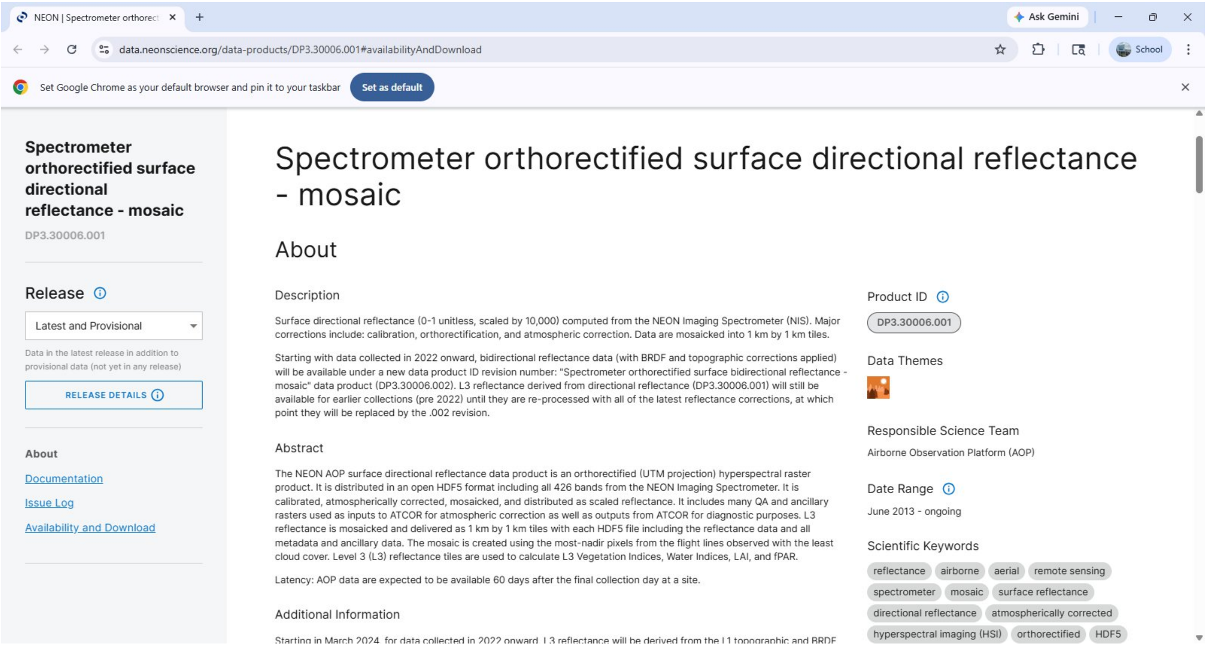
Abstract

Albedo, the ratio of a surface's reflected energy to its incident energy, is an important measurement for characterizing earth system energy balance. Light and dark surfaces correspond to high and low albedo, respectively. An opaque surface's difference in energy reflected as compared to the energy incident on it is absorbed by the surface, increasing its temperature (Sabins, Jr., 1978). Albedo values depend on wavelength, illumination sources and geometry, sensor viewing geometry, reflectance as a function of angle and wavelength, as well as the scattering, absorbing, and re-radiating effects of the atmosphere. These factors are modeled/accounted for to best approximate a bi-hemispherical reflectance as would be measured in a laboratory setting. To this end, the wavelength-integrated surface reflectance, weighted with the global flux on the ground, is produced as the best practically achievable albedo measurement (Richter & Schlapfer, 2017). L3 Albedo is distributed in 1 km by 1 km tiles created by taking the most-nadir pixels from the flight lines acquired in the best weather conditions.

Latency: AOP data will be available 60 days after the final collection day at a site. Note: Data for airborne collections in 2022 and 2023 are currently being added to the data portal under a new revision number (DP3.30011.002) since they are derived from BRDF corrected reflectance data, so will not be available here. Albedo derived from directional reflectance will still be available for earlier collections (pre 2022) until they are re-processed with all of the latest reflectance corrections.

Additional Information

Starting in March 2024, for data collected in 2022 onward, L3 albedo data will be derived from topographic and BRDF corrected reflectance data (DP1.30006.002). For 2022 and 2023, albedo data can be found under "Albedo - spectrometer - bidirectional mosaic" (DP3.30011.002). AOP spectrometer data collected between 2013-2021 will be backprocessed to apply the BRDF and topographic corrections, so as these data become available under the new data product ID (DPID) revision, they will be removed from this DPID.



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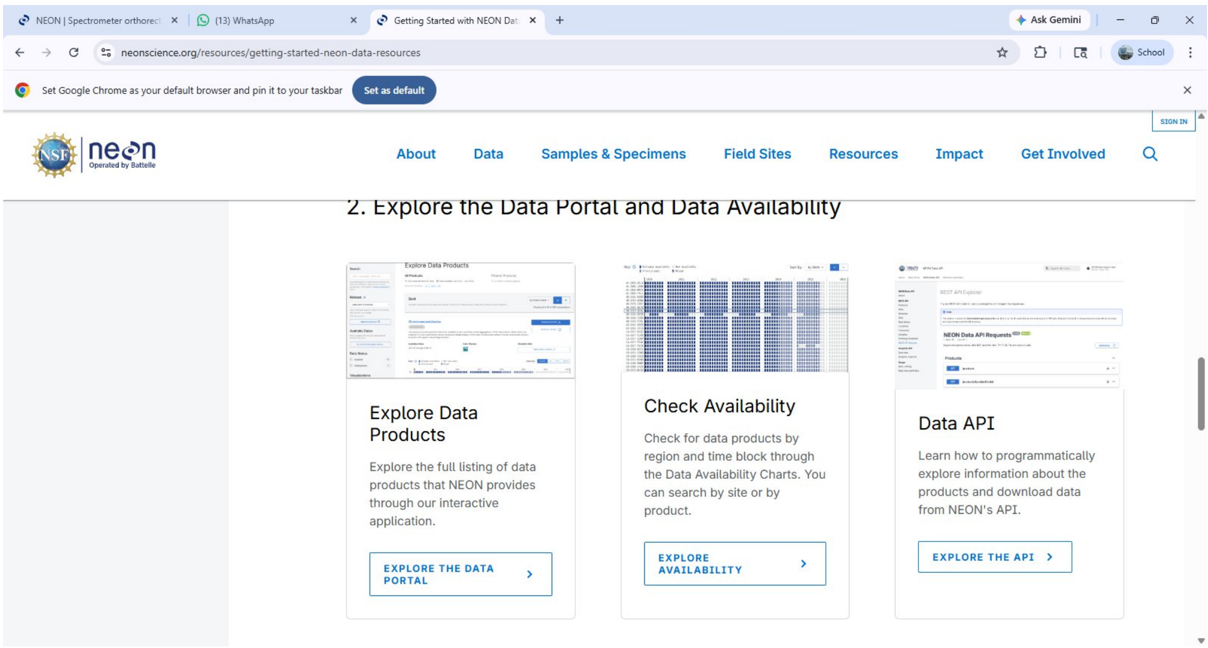


Study Description

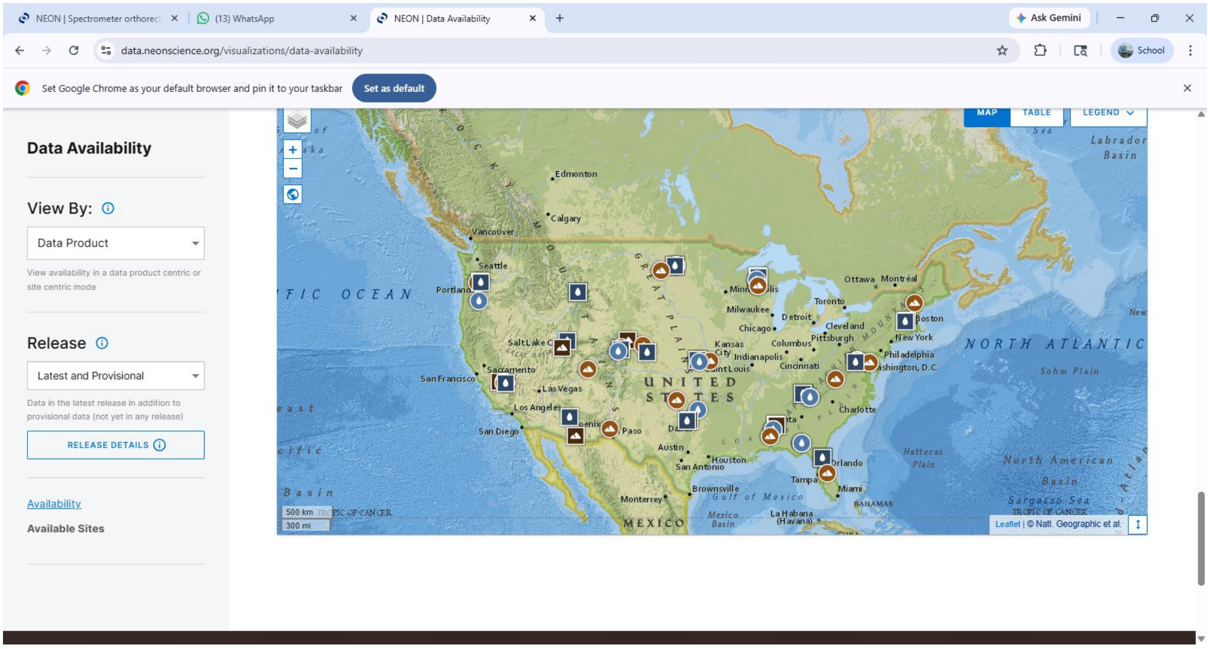
NEON AOP data are planned for yearly collects at all NEON sites at 90% of maximum greenness or greater. Coverage is planned to include at least 95% of NIS Tower Airshed area as well as at least 80% of a minimum 10km x 10 km box around that. All acquisitions are subject to change due to weather conditions as well as program planning changes.

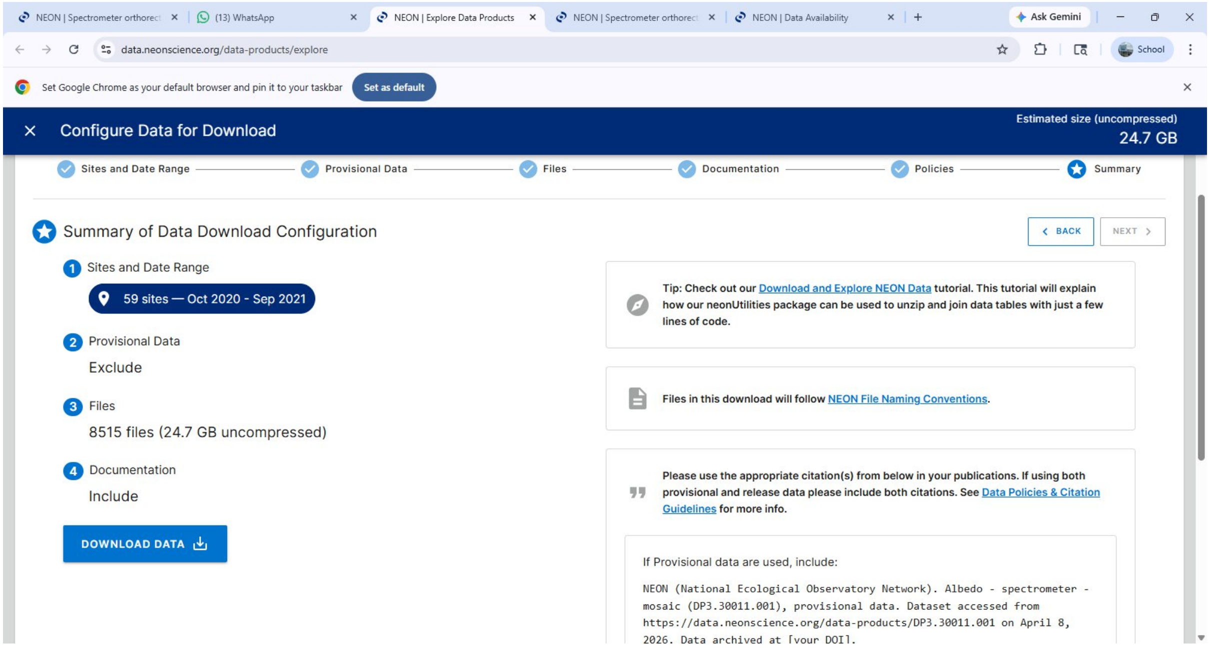
Design Description

Albedo data products are derived from NEON Airborne Observation Platform (AOP) Imaging Spectrometer (NIS) data collected in North-South oriented flight lines to reduce BRDF effects. These data are processed to orthorectified directional surface reflectance and then to Albedo as best practically approximated by wavelength-integrated surface reflectance, weighted with the global flux on the ground. L3 Albedo is distributed in 1 km by 1 km tiles at a 1 m spatial resolution. Filenames contain the coordinate of the lower-left (south west) corner of the tile.

Instrumentation

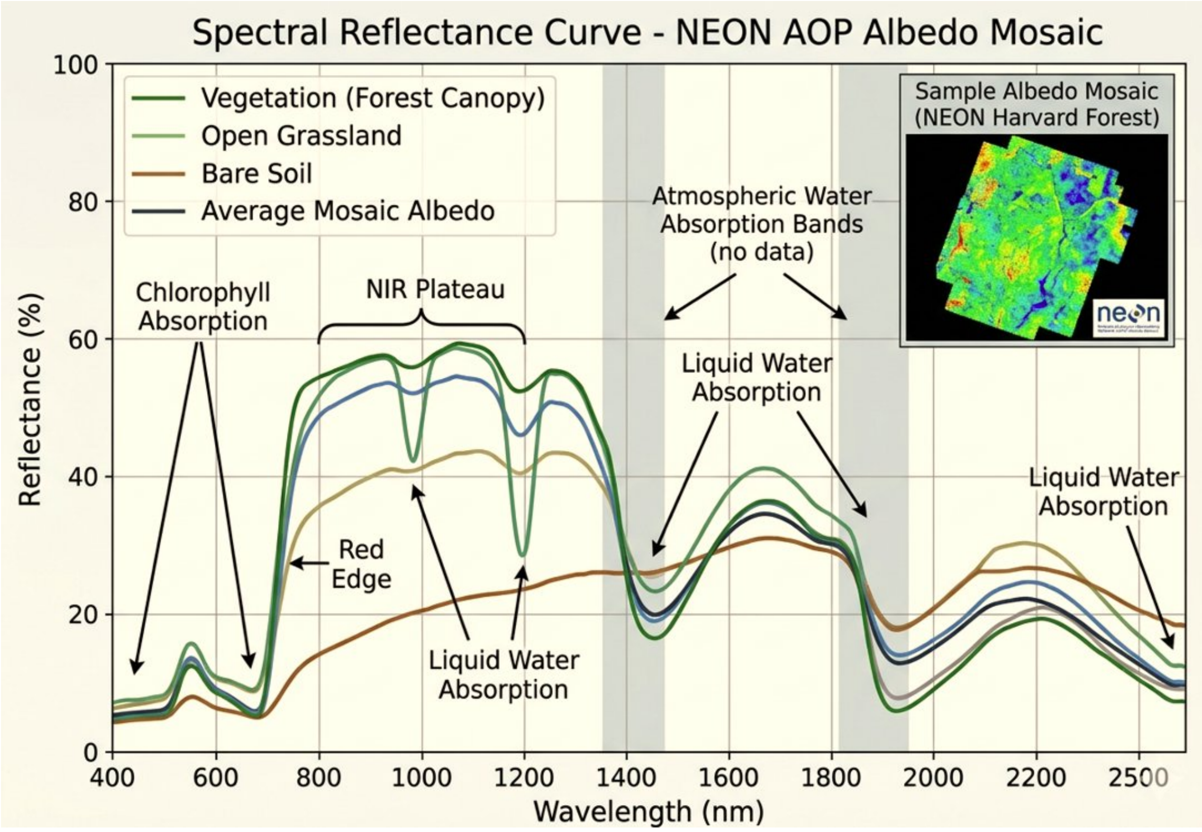
NEON Airborne Observation Platform (AOP) Imaging Spectrometer (NIS) - NASA/JPL AVIRIS-NG

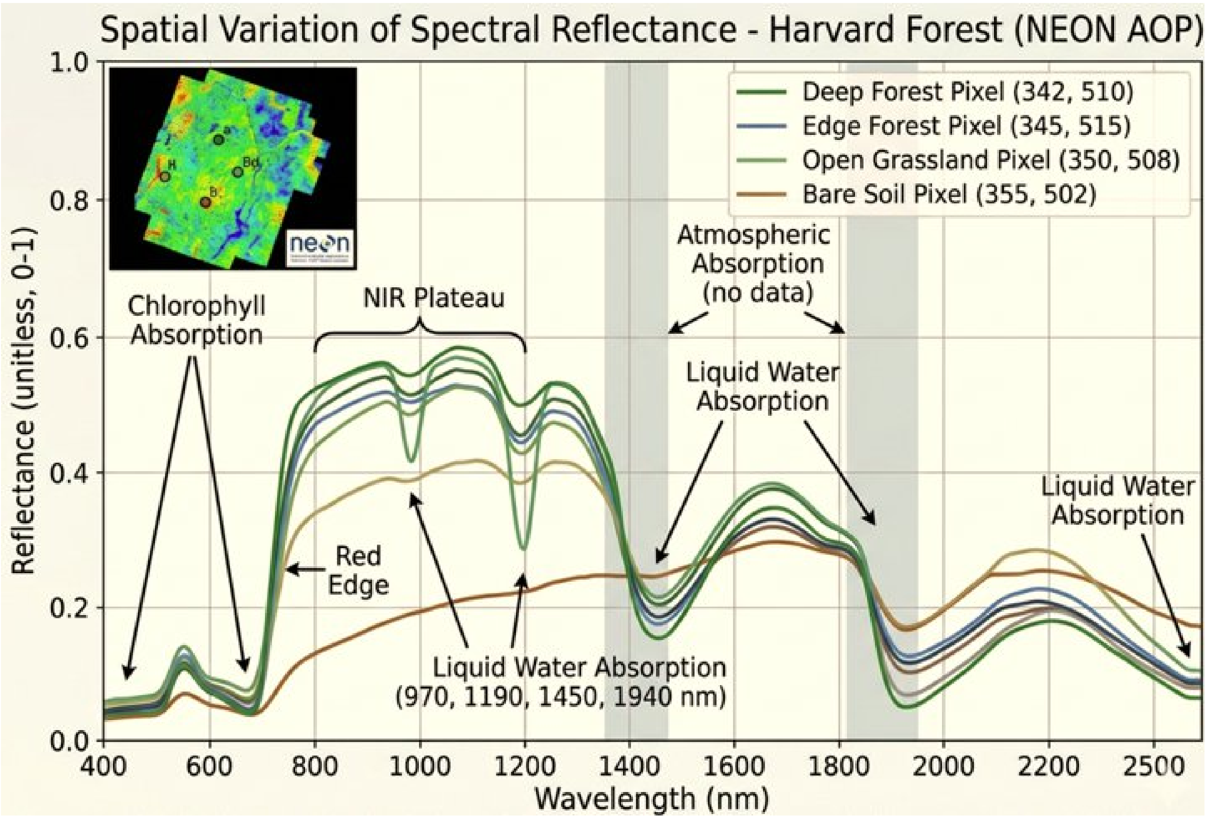




PYTHON CODE TO GENERATE SPECTRAL REFLECTANCE CURVE:

```
import matplotlib.pyplot as plt

import numpy as np


# Example Data (Wavelengths in nm, Reflectance from 0.0 to 1.0)

wavelengths = np.linspace(400, 2500, 100)

# Simple mock curve simulating vegetation (low in visible, high in NIR)

reflectance = [0.05 if w < 700 else 0.45 for w in wavelengths]


plt.figure(figsize=(10, 6))

plt.plot(wavelengths, reflectance, color='black', linewidth=2)


plt.title('Spectral Reflectance Curve', fontsize=14)

plt.xlabel('Wavelength (nm)', fontsize=12)

plt.ylabel('Reflectance (0-1)', fontsize=12)

plt.grid(True, linestyle='--', alpha=0.7)

plt.show()
```





Spectral reflectance curve

Conclusion:

Albedo - spectrometer - mosaic (DP3.30011.001) Measurement The ratio of solar energy reflected off of the earth's surface to total incident energy in the 0.4 to 2.5 micron wavelength range and modeled to best account for the many influencing factors such as reflectance angle /wavelength, solar angle, and scattering, absorbing, & re-radiating effects of atmospheric conditions. Albedo data values range from 0 to 1. All AOP L3 data are distributed in 1 km by 1 km tiles which include the Easting and Northing coordinates (in that order) of their southwest corner in their file names. Collection methodology NEON AOP instruments, including the NEON Imaging Spectrometer (NIS), are flown at 1000 m above ground level at 100 knots in flight lines that have 37% overlap. This results in NIS collections at 1 m spatial resolution and 5 nm spectral resolution in over 420 bands from 380 to 2500 nm. All AOP data are acquired in individual flight lines (typically 600m wide by 5 to 20 km long). NEON sites are flown once per year with a target of 90% of maximum greenness or higher and at a minimum of 3 of every 4 years. Flight coverage is a minimum of 10 km by 10 km and covers both NEON tower and observational sampling sites. Aquatic sites, as well as the terrestrial sites in Hawaii and Puerto Rico, are flown in the same fashion, but at reduced frequency. For information about disturbances, land management activities, and other incidents that may impact data at NEON sites, see the Site management and event reporting (DP1.10111.001) data product. Maintenance and calibration Calibration activities are conducted on the NIS every winter between flight seasons in the NEON AOP lab where it measures known radiance sources in a controlled environment. Calibration/validation flights are flown at the beginning and end of every flight season for positional and timing accuracy as well as measuring radiance and reflectance over well-characterized, homogenous targets. In each domain collected, an in-situ calibration is conducted where the NIS records radiance from a known portable mercury lamp positioned directly under the sensor. Finally, at the start and end of each flight line flown, a set of on-board calibrations are made. These calibration activities allow any variances in the spectral response of the focal plane to be characterized, recorded, and tracked. Data processing and derivation Each NIS flight line is processed from raw L0 DN to L1 radiance using the lab calibration information. Each radiance flight line is then orthorectified using a digital surface models from the co-collected lidar data. The orthorectified radiance flight lines are then processed to directional surface reflectance using the atmospheric correction and modeling package ATCOR. ATCOR is also used to integrate the reflectance values across the albedo wavelength range and ratio against the modeled incident radiance for the flightline Page 2 of 2 acquisition time and location, producing the final albedo values written out to GeoTiff files for individual flight lines. The flightline GeoTiff files at a given site are then merged and distributed in 1 km by 1 km tiles where pixels are selected from flightlines with the highest quality cloud conditions at collect and closest proximity to nadir. Data quality Downloaded data include a quality report for individual flights, as well the culmination of flights over a site, describing cloud conditions, positioning errors, parameters used for atmospheric corrections, and other data quality considerations made during flight collection and processing of data. Flightlines are each assigned color-coded quality flags indicating cloud cover (green 50%). Pixels missing data are coded -9999. Please note that quality checks are comprehensive but not exhaustive; therefore, unknown data quality issues may exist. Users are advised to evaluate quality of the data as relevant to the scientific research question being addressed, perform data review and post-processing prior to analysis, and use the data quality information and issue logs included in download packages to aid interpretation.